

Congestion – Restrict to competitively awarded projects

2026-04-09

- $Treat_i = 1$ for contractors that hold at least one small project over the sample horizon.
- $Treat_i = 0$ for contractors that hold only large projects.
- For treated group, focus only on new projects/budget/etc. acquired for small projects.
- For the control group, by construction, changes will account for new projects/tasks/budget acquired for large projects

Table 1: Congestion and the QuickPay reform

	$NumProjects_{kt}$		$TotalBudget_{kt}$ (millions)		$NumTasks_{kt}$		$TaskVariability_{kt}$	
	I	II	III	IV	V	VI	VII	VIII
Constant	5.15*** (0.44)		4.11*** (0.57)		1.43*** (0.02)		0.13*** (0.00)	
$Treat_k$	-1.02** (0.47)	-1.01** (0.47)	-2.86*** (0.58)	-2.84*** (0.58)	-0.02 (0.03)	-0.02 (0.03)	0.05*** (0.01)	0.05*** (0.01)
$Post_t$	0.54*** (0.20)		1.97*** (0.28)		0.05*** (0.01)		-0.02*** (0.00)	
$Treat_k \times Post_t$	-0.01 (0.21)	-0.02 (0.21)	-1.16*** (0.28)	-1.18*** (0.28)	0.02 (0.02)	0.02 (0.02)	0.01*** (0.01)	0.01*** (0.01)
Time FE	No	Yes	No	Yes	No	Yes	No	Yes
Observations	77,547	77,547	77,547	77,547	77,547	77,547	60,717	60,717
R ²	0.001	0.002	0.02	0.02	0.0007	0.002	0.009	0.02

Clustered (Contractor) standard-errors in parentheses

Signif. Codes: ***, 0.01, **, 0.05, *, 0.1

Sample restricted to competitively awarded projects. Treat is an indicator that equals one for contractors that hold at least one small project during the observation horizon. Control group constitutes contractors who hold only large projects during the observation horizon.

Table 2: Congestion and the QuickPay reform

	$TaskVariability_{kt}$		
	I	II	III
Constant	0.13*** (0.00)		
$Treat_k$	0.05*** (0.01)	0.05*** (0.01)	
$Post_t$	-0.02*** (0.00)		
$Post_t \times Treat_k$	0.01*** (0.01)	0.01*** (0.01)	0.01** (0.01)
Time FE	No	Yes	Yes
Contractor FE	No	No	Yes
Observations	60,717	60,717	58,155
R ²	0.009	0.02	0.49

Clustered (Contractor) standard-errors in parentheses

Signif. Codes: ***, 0.01, **, 0.05, *, 0.1

Sample restricted to competitively awarded projects. Treat is an indicator that equals one for contractors that hold at least one small project during the observation horizon. Control group constitutes contractors who hold only large projects during the observation horizon.

Table 3: Congestion and the QuickPay reform

	<i>NumProjects_{kt}</i>			<i>TotalBudget_{kt}</i> (millions)			<i>NumTasks_{kt}</i>			<i>TaskVariability_{kt}</i>		
	I	II	III	IV	V	VI	VII	VIII	IX	X	XI	XII
Constant	5.15*** (0.44)			4.11*** (0.57)			1.43*** (0.02)			0.13*** (0.00)		
<i>Treat_k</i>	-1.02** (0.47)	-1.01** (0.47)		-2.86*** (0.58)	-2.84*** (0.58)		-0.02 (0.03)	-0.02 (0.03)		0.05*** (0.01)	0.05*** (0.01)	
<i>Post_t</i>	0.54*** (0.20)			1.97*** (0.28)			0.05*** (0.01)			-0.02*** (0.00)		
<i>Post_t × Treat_k</i>	-0.01 (0.21)	-0.02 (0.21)	-0.49* (0.27)	-1.16*** (0.28)	-1.18*** (0.28)	-1.93*** (0.31)	0.02 (0.02)	0.02 (0.02)	-0.02 (0.02)	0.01*** (0.01)	0.01*** (0.01)	0.01** (0.01)
Time FE	No	Yes	Yes	No	Yes	Yes	No	Yes	Yes	No	Yes	Yes
Contractor FE	No	No	Yes	No	No	Yes	No	No	Yes	No	No	Yes
Observations	77,547	77,547	72,308	77,547	77,547	72,308	77,547	77,547	72,308	60,717	60,717	58,155
R ²	0.001	0.002	0.83	0.02	0.02	0.80	0.0007	0.002	0.81	0.009	0.02	0.49

Clustered (Contractor) standard-errors in parentheses

Signif. Codes: ***, 0.01, **, 0.05, *, 0.1

Sample restricted to competitively awarded projects. Treat is an indicator that equals one for contractors that hold at least one small project during the observation horizon. Control group constitutes contractors who hold only large projects during the observation horizon.

Table 4: Similarity score and the QuickPay reform

	<i>PercentDelayRate_{it}</i>			
	I	II	III	IV
<i>TaskVariability_{kt}</i>	5.64*** (0.58)	2.97*** (0.58)	3.16*** (0.58)	2.69*** (0.75)
Project stage	2.82*** (0.03)	2.73*** (0.03)	2.71*** (0.03)	2.37*** (0.03)
Initial duration	-8.06*** (0.12)	-7.74*** (0.13)	-7.69*** (0.13)	-7.33*** (0.14)
Initial budget	0.93*** (0.02)	0.46*** (0.03)	0.44*** (0.03)	0.34*** (0.03)
Number of offers	-0.07*** (0.01)	-0.05*** (0.01)	-0.04*** (0.01)	-0.03* (0.01)
<i>Post_t × Treat_i</i>	0.04 (0.12)	0.50*** (0.12)	0.49*** (0.12)	0.78*** (0.19)
<i>TaskVariability_{kt} × Post_t</i>	-2.58*** (0.77)	-0.78 (0.77)	-0.87 (0.77)	0.45 (0.86)
<i>TaskVariability_{kt} × Treat_i</i>	-8.17*** (0.60)	-3.29*** (0.59)	-3.38*** (0.59)	-1.95** (0.82)
<i>Post_t × Initial duration</i>	0.92*** (0.16)	0.55*** (0.16)	0.52*** (0.16)	0.44*** (0.16)
<i>Post_t × Initial budget</i>	0.04 (0.03)	-0.01 (0.03)	0.00 (0.03)	0.04 (0.04)
<i>Post_t × Number of offers</i>	0.09*** (0.02)	0.06*** (0.01)	0.06*** (0.01)	0.06*** (0.02)
<i>TaskVariability_{kt} × Post_t × Treat_i</i>	4.15*** (0.84)	1.98** (0.84)	2.01** (0.84)	1.17 (0.97)
Time FE	Yes	Yes	Yes	Yes
Task FE	No	Yes	Yes	Yes
Industry FE	No	No	Yes	Yes
Contractor FE	No	No	No	Yes
Observations	165,541	165,431	165,426	163,188
R ²	0.18	0.22	0.22	0.28

Clustered (Project ID) standard-errors in parentheses

Signif. Codes: ***, 0.01, **, 0.05, *, 0.1

Sample restricted to competitively awarded projects. Treat is an indicator that equals one for small projects. Control group constitutes projects belonging to contractors who hold only large projects during the observation horizon.

Effect on \$TaskVariability_{kt}\$

